Stefano Longari

Curriculum Vitae

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Last updated: April 2026

Contents

Overview

I am a fixed-term researcher (RTDa) at Politecnico di Milano, Department of Electronics, Information and Bioengineering (DEIB), where I work within the NECSTLab system security research group. My research focuses on identifying and mitigating threats to cyber-physical systems and critical infrastructures, spanning automotive networks, satellite systems, industrial robots, and smart infrastructure, with an emphasis on intrusion detection, threat modeling, and the security implications of machine learning. A secondary research direction addresses social engineering and the evolving role of AI in enabling and defending against it.

My work has appeared in top-tier security venues, including IEEE S&P (to appear, 2026), USENIX Security (2025), ACM CCS (2022, Best Paper Honorable Mention), and PETS (2025), as well as in leading IEEE and ACM transactions. I am the author of 9 journal papers (7 in Q1-ranked journals), 22 peer-reviewed conference papers (3 at A* venues), and 2 patents.

I serve as course instructor for "Human and Physical Aspects of Security" (jointly offered to Politecnico and Bocconi students), covering threat modeling, cyber-physical systems security, and social engineering, and I coordinated the "Security Specialist" specialization degree for CEFRIEL and Politecnico di Milano. I act as a reviewer for Q1 journals including IEEE T-IFS, IEEE TDSC, and ACM Computing Surveys, and have served on the program committees of ACSAC, DIMVA, and multiple workshops co-located with IEEE S&P and ACM CCS. I currently advise five Ph.D. students (including one co-advised) and have advised or co-advised more than fifty master's theses.

Current Role	Publications	Google Scholar	Scopus	Patents
RTDa, PoliMi (since May 2023)	9 journals (7 Q1) 22 conf. papers (3 A*)	h-index 12 508 citations	h-index 9 295 citations	2 (1 international)

Note: citation metrics should be verified against live profiles before submission.

Research Interests

My research began during my Ph.D. with a focus on automotive cybersecurity, specifically on the Controller Area Network (CAN) protocol. This work produced real-world attack datasets, novel threat demonstrations at the data-link layer, and a family of machine learning-based intrusion detection systems. The CANflict paper, which demonstrated a new class of peripheral-conflict attacks on CAN, received a Best Paper Honorable Mention at ACM CCS 2022. More recently, I have investigated the robustness of these detection systems against adversarial manipulation.

Building on this foundation, I have expanded into the broader domain of **cyber-physical systems security**. Current lines of work include: (i) *satellite and space systems*, where I acted as a local PI for the SOCRATE cyber range project (funded by ASI) and an industrial contract with D-Orbit on intra-satellite IDS, with publications at the 3S conference and multiple ongoing theses; (ii) *industrial control systems*, including TEE-based attack detection for robotic controllers and novel attacks on PROFIbus/PROFIsafe protocols; (iii) *smart infrastructure*, encompassing false data injection detection in water systems and adaptive control under denial-of-service for regulated dams; and (iv) *e-vehicle charging*, where I advise a Ph.D. student and have published security analyses of charging infrastructure.

A growing research direction concerns the **security of machine learning pipelines**. Our S&P 2026 paper uncovered six 0-day vulnerabilities in ML model-sharing frameworks and hubs. Related work addresses privacy leakage in federated tree-based systems, decentralized federated learning security, and the limitations of large language models for automated vulnerability detection.

Finally, I maintain a research interest in **social engineering and AI-driven threats**, including the systematic study of how LLMs can be leveraged for spear phishing and the broader impact of AI on social engineering over the past decade. This line informs both my teaching and my consulting and media work for RAI and WindTre.

Experience

Junior Researcher (RTDa) — May 2023 — Present
Politecnico di Milano (Milan, Italy)
Research topic: Threat modeling, offensive, and defensive measures for novel cyber-physical systems.
Research group: NECSTLab, System security research group.

Postdoctoral Researcher — Jun 2021 — Apr 2023
Politecnico di Milano (Milan, Italy)
Research topic: Offensive and defensive security techniques for cyber-physical systems.
Research group: NECSTLab, System security research group.

Research Internship — Oct 2019 — Apr 2020
ESCRYPT GmbH (Stuttgart, Germany)
Research topic: Development of new security techniques for automotive on-board CAN-connected devices.

Education

Ph.D. in Information Technology — Jun 2021
Politecnico di Milano
Dissertation: *On the security of connected automotive systems*

M.Sc. in Computer Science and Engineering — Apr 2018
Politecnico di Milano
Dissertation: *On the security of connected vehicles*

B.Sc. in Computer Science and Engineering — Sep 2015
Politecnico di Milano

Publications

Productivity and Impact Metrics

Author/Co-author of 9 scientific publications in peer-reviewed journals, including 7 top-ranked Q1 journals (SCIMAGO), and 22 peer-reviewed conference papers, including 3 at top-ranked A* security conferences (ACM CCS, USENIX Security, IEEE S&P). Two conference papers received awards: Best Paper Honorable Mention at

ACM CCS 2022 and Best Paper Runner Up at VehicleSec 2024. Author of 2 patents. First or corresponding author on 6 of 9 journal papers and 8 of 22 conference papers.

Based on Google Scholar: h-index 12, 508 citations, 35 entries.
Based on Scopus: h-index 9, 295 citations, 26 entries.

Peer-reviewed Journals

9. Digregorio, G., Saputelli, E., **Longari, S.**, Carminati, M., & Zanero, S. (2025). Swarm: A Distributed Ledger-based Framework to Enhance Air Traffic Control Security Using ADS-B Protocol. *ACM Transactions on Privacy and Security*, 28(4), 44:1–44:35.
8. **Longari, S.**, Cerracchio, P., Carminati, M., & Zanero, S. (2025). Assessing the Resilience of Automotive Intrusion Detection Systems to Adversarial Manipulation. *ACM Transactions on Cyber-Physical Systems*, 9(3), 31:1–31:27.
7. Barelli, R., D'Onghia, M., & **Longari, S.** (2025). Toward Secure Electronic Voting: a Survey on E-Voting Systems and Attacks. *IEEE Access*, 13, 89600–89626.
6. **Longari, S.**, Jannone, J., Polino, M., Carminati, M., Zanchettin, A. M., Tanelli, M., & Zanero, S. (2025). Janus: A Trusted Execution Environment Approach for Attack Detection in Industrial Robot Controllers. *IEEE Transactions on Emerging Topics in Computing*, 13(1), 185–195.
5. **Longari, S.**, Pozone, A., Leoni, J., Polino, M., Carminati, M., Tanelli, M., & Zanero, S. (2023). CyFence: Securing Cyber-physical Controllers Via Trusted Execution Environment. *IEEE Transactions on Emerging Topics in Computing*, 11(4), 1010–1022.
4. Nichelini, A., Pozzoli, C. A., **Longari, S.**, Carminati, M., & Zanero, S. (2023). Canova: a hybrid intrusion detection framework based on automatic signal classification for CAN. *Computers & Security*, 128, 103166.
3. Maffiola, D., **Longari, S.**, Carminati, M., Tanelli, M., & Zanero, S. (2021). GOLIATH: A Decentralized Framework for Data Collection in Intelligent Transportation Systems. *IEEE Transactions on Intelligent Transportation Systems*, 23(8), 13006–13016.
2. **Longari, S.**, Valcarcel, D. H. N., Zago, M., Carminati, M., & Zanero, S. (2020). CANnolo: An anomaly detection system based on LSTM autoencoders for controller area network. *IEEE Transactions on Network and Service Management*, 18(2), 1913–1924.
1. Zago, M., **Longari, S.**, Tricarico, A., Carminati, M., Pérez, M. G., Pérez, G. M., & Zanero, S. (2020). ReCAN—Dataset for reverse engineering of Controller Area Networks. *Data in brief*, 29, 105149.

Peer-reviewed Conference Proceedings

22. Digregorio, G., Di Gennaro, M., Zanero, S., **Longari, S.**, & Carminati, M. (2026). When Secure Isn't: Assessing the Security of Machine Learning Model Sharing. *IEEE Symposium on Security and Privacy (S&P 2026)*. **To appear.**
21. Büchel, M., Paladini, T., **Longari, S.**, Carminati, M., Zanero, S., Binyamini, H., Engelberg, G., Klein, D., Guizzardi, G., Caselli, M., Continella, A., van Steen, M., Peter, A., & van Ede, T. (2025). SoK: Automated TTP Extraction from CTI Reports—Are We There Yet? *34th USENIX Security Symposium (USENIX Security 2025)*.
20. Santorsola, A., Mammone, D., **Longari, S.**, Topputo, F., & Mergè, M. (2025). An End-to-End GEO Satellite Links Simulation Framework for Cyber Range Applications. *2025 Security for Space Systems (3S)*. IEEE.
19. Di Gennaro, M., De Lucia, G., **Longari, S.**, Zanero, S., & Carminati, M. (2025). TimberStrike: Dataset Reconstruction Attack Revealing Privacy Leakage in Federated Tree-Based Systems. *25th Privacy Enhancing Technologies Symposium (PETS 2025)*, 2025(4), 566–584.
18. Digregorio, G., Bleggi, F., Caroli, F., Carminati, M., Zanero, S. & **Longari, S.** (2025). Poster: FedBlockParadox—A Framework for Simulating and Securing Decentralized Federated Learning. *DIMVA 2025* (pp. 323–328).
17. Mammone, D., Bossi, L., Carminati, M., Zanero, S., & **Longari, S.** (2025). Linux hurt itself in its confusion! Exploiting Out-of-Memory Killer for Confusion Attacks via Heuristic Manipulation. *DIMVA 2025* (pp. 195–215).
16. Panebianco, F., Isgrò, A., **Longari, S.**, Zanero, S., & Carminati, M. (2025). Guessing As A Service: Large Language Models Are Not Yet Ready For Vulnerability Detection. *ITASEC & SERICS 2025*.
15. Balossini, M., Carminati, M., Zanero, S., & **Longari, S.** (2025). Micro-Mobility Security: A Holistic Approach via Mobile App Analysis. *ITASEC & SERICS 2025*.
14. Abbasi, S. M., & **Longari, S.** (2025). CANPak: An Intrusion Detection System against Error Frame Attacks for Controller Area Network. *ITASEC & SERICS 2025*.

13. Giannubilo, D., Giorgeschi, T., Carminati, M., Zanero, S., & **Longari, S.** (2025). A Deep Learning Approach for False Data Injection Attacks Detection in Smart Water Infrastructure. *ITASEC & SERICS 2025*.

12. **Longari, S.**, Galletti, G., Holle, J., & Zanero, S. (2025). CANter: data-link layer detection of drop-and-spoof attacks on CAN and CAN FD. *ITASEC & SERICS 2025*.

11. Amico, A., Apicella, V., Bianchini, D., et al. (2024). BOTQUAS: Blockchain-based Solutions for Trustworthy Data Sharing in Sustainable and Circular Economy. *EUROMICRO-SEAA 2024*.

10. Cestari, R. G., **Longari, S.**, Zanero, S., & Formentin, S. (2024). Model Predictive Control with adaptive resilience for Denial-of-Service Attacks mitigation on a Regulated Dam. *IEEE 63rd Conference on Decision and Control (CDC)*.

9. Digregorio, G., Cainazzo, E., **Longari, S.**, Carminati, M., & Zanero, S. (2024). Evaluating the Impact of Privacy-Preserving Federated Learning on CAN Intrusion Detection. *IEEE VTC Spring 2024*.

8. Cerracchio, P., **Longari, S.**, Carminati, M., & Zanero, S. (2024). Investigating the Impact of Evasion Attacks Against Automotive Intrusion Detection Systems. *VehicleSec 2024*.

7. Marazzi, M., **Longari, S.**, Carminati, M., & Zanero, S. (2024). Securing LiDAR Communication through Watermark-based Tampering Detection. *VehicleSec 2024*.

6. **Longari, S.**, Pozzoli, C. A., Nichelini, A., Carminati, M., & Zanero, S. (2023). Candito: improving payload-based detection of attacks on controller area networks. *CSCML 2023* (pp. 135–150). Springer.

5. **Longari, S.**, Noseda, F., Carminati, M., & Zanero, S. (2023). Evaluating the Robustness of Automotive Intrusion Detection Systems Against Evasion Attacks. *CSCML 2023* (pp. 337–352). Springer.

4. Avanzi, D., **Longari, S.**, Polino, M., Carminati, M., Zanchettin, A. M., Tanelli, M., & Zanero, S. (2023). Task Aware Intrusion Detection for Industrial Robots. *ITASEC 2023* (pp. 1–16). CEUR.

3. de Faveri Tron, A., **Longari, S.**, Carminati, M., Polino, M., & Zanero, S. (2022). CANflict: Exploiting Peripheral Conflicts for Data-Link Layer Attacks on Automotive Networks. *ACM CCS 2022* (pp. 711–723).

2. **Longari, S.**, Penco, M., Carminati, M., & Zanero, S. (2019). CopyCAN: An error-handling protocol based intrusion detection system for controller area network. *ACM CPS-SPC 2019* (pp. 39–50).

1. **Longari, S.**, Cannizzo, A., Carminati, M., & Zanero, S. (2019). A secure-by-design framework for automotive on-board network risk analysis. *IEEE VNC 2019* (pp. 1–8).

Awards

- Best Paper Runner Up, *Securing LiDAR Communication through Watermark-based Tampering Detection*. VehicleSec 2024.
- Best Paper Honorable Mention, *CANflict: Exploiting Peripheral Conflicts for Data-Link Layer Attacks on Automotive Networks*. ACM CCS 2022.

Patents

2. Computer-implemented real-time control system for controlling a physical system or device. International ref: WO2023002321A1. Italian pub. no: IT202100018998A1. **Longari, S.**, Pozone, A., Tanelli, M., Zanero, S. Published 2023-01-26.

1. Verfahren zum Erkennen eines Angriffs auf einen zu sichernden Busteilnehmer, Überwachungseinheit und Bussystem. Reg. no: DE 102022207911.6. **Longari, S.**, Holle, J. Registered 2022-08-01.

Funded Projects

European Projects

- **AHEAD**: AI-informed Holistic EVs integration Approaches for Distribution grids — 2024–2028
  Role: Task Leader.
- **SARIL**: Sustainability And Resilience for Infrastructure and Logistics networks — 2023–2026
  Role: Participant.
- **KEEPER**: Key Code based on Nanomaterials to Protect Services and Products — 2024–2027
  Role: Participant.

Other Funded Projects

- **SOCRATE**: Satellite Operations Cyber Range Evaluation — 2023–2025
  Funding source: Agenzia Spaziale Italiana. Role: Local PI.

- **MICS**: Made in Italy Circolare e Sostenibile 2023–2025
  Funding source: PNRR. Role: Participant.
- **ACN Cybersecurity Ph.D. position**: funded position on satellite cybersecurity 2026–2028
  Funding source: Agenzia per la Cybersicurezza Nazionale. Role: Proponent.

Industrial Contracts

- **Satellite Cybersecurity**: development of Intrusion Detection Solutions for intra-satellite networks 2024–2025
  Funding source: D-Orbit. Role: PI.

Academic Roles

Specialization Degree Coordinator

Security Specialist specialization degree ("Master universitario di primo livello"), CEFRIEL and Politecnico di Milano. ∼20 students, 2-year program. 2022 — Present.

Other Academic Roles

- Academic Advisor for Politecnico's Career Service 2025 — Present
  Handles industrial stages for Cyber Risk Strategy and Governance master students.
- Academic Cybersecurity Area Referent for MADE Competence Center 2025 — Present
- CPDS Member for the Cyber Risk Strategy and Governance Master 2025 — Present
- Examination Board Chair, Computer Science and Engineering Master, Politecnico di Milano 2024 — Present
- Examination Board Member, Computer Science and Engineering, Politecnico di Milano 2022 — Present
- Examination Board Member, Cyber Risk Strategy and Governance, Bocconi and Politecnico 2023 — Present

Events and Conferences Organization

OWASP Italy Day 2023 — Organizers: OWASP Italy and Politecnico di Milano.

Program Committee

- Annual Computer Security Applications Conference, ACSAC 2025
- Italian conference on Cybersecurity, ITASEC 2024–2025
- Conference on Detection of Intrusions and Malware & Vulnerability Assessment, DIMVA 2024
- Workshop on Re-design Industrial Control Systems with Security, RICSS (Euro S&P) 2023
- Cyber-Physical System Security Workshop, CPSS (ASIACCS) 2023
- Workshop on Automotive Cybersecurity, ACSW (EURO S&P) 2022–2025
- Joint Workshop on CPS & IoT Security and Privacy, CPSIoTSec (ACM CCS) 2022
- Int. Conference on Security for IT and Communications, SecITC 2022

Subreviewer

- IEEE Symposium on Security and Privacy, IEEE S&P 2023–2025
- ACM Conference on Computer and Communications Security, ACM CCS 2023–2024

Reviewer for Q1 Journals and Transactions

- ACM Transactions on Cyber-Physical Systems, ACM TCPS 2025
- IEEE Internet of Things Journal, IEEE IoT 2022–2025
- Elsevier Computers and Security, COSE 2019–2025
- ACM Computing Surveys, ACM CSUR 2024
- IEEE Transactions on Automation Science and Engineering, IEEE T-ASE 2024
- Elsevier J. of Parallel and Distributed Computing, JPDC 2024
- IEEE Transactions on Intelligent Transportation Systems, IEEE T-ITS 2023
- IEEE Transactions on Aerospace and Electronic Systems, IEEE TAES 2022–2023
- IEEE Transactions on Dependable and Secure Computing, IEEE TDSC 2022
- IEEE Transactions on Industrial Informatics, IEEE TII 2021–2022
- IEEE Transactions on Information Forensics and Security, IEEE T-IFS 2021
- IEEE Transactions on Emerging Topics in Computing, IEEE TETC 2018

Speaker Engagements

Invited Talks

- REDDER Level Up 2025 — *Il lato oscuro delle intercettazioni*
- Culturstrike 2025 — *Cybersecurity - AI ed il fattore umano*
- REDDER Level Up 2024 — *L'anello debole: l'IA punta sugli esseri umani*
- Bosch Next Event 2024 — *Challenges in Industrial Cybersecurity*
- Hromatka Group 2023 — *Risk and Industrial Security*
- HackInBo 2022 — *The CAN Link-Layer, or how we implemented a broken protocol, and can we fix it?*
- Automotive Security Research Group 2021 — *CAN Error Handling Attacks and Countermeasures*
- Hardwear.io 2019 — *It's easier to break than to patch: a stealthy DoS attack against CAN*
- Infosek conference 2018 — *Automotive Security*

Paper Presentations

- ITASEC 2025: CANPak; A Deep Learning Approach for FDI Attacks Detection in Smart Water Infrastructure.
- ITASEC 2025: CANter.
- VehicleSec 2024: Investigating the Impact of Evasion Attacks Against Automotive IDS; Securing LiDAR Communication.
- CSCML 2023: Candito; Evaluating the robustness of automotive IDS against evasion attacks.
- IEEE VNC 2019: A secure-by-design framework for automotive on-board network risk analysis.
- ACM CPS-SPC 2019: CopyCAN.

Advisor Activities

Ph.D. Thesis Advisor

- (co-advisor) Cyber threat identification and mitigation in the New Space Era. Bruzzese Luigi 2026 — Present
- Artificial intelligence for automated cyber threat detection and analysis. Bossi Lorenzo 2025 — Present
- On the security of e-vehicle charging infrastructure. Balossini Marco 2024 — Present
- Security of Cyber-Physical Systems. Mammone Daniele 2023 — Present
- Offensive and Defensive Cybersecurity for Critical Infrastructures. Digregorio Gabriele 2023 — Present

Master Thesis Advisor (selection)

10. X-rated Compliance Theater: An Empirical Evaluation of European Age Verification Systems in Adult Websites. Lavermicocca, Simone, 2026.
9. Artificial Scams: On the risks of Fully Agentic Spear Phishing. Maroni, Manuela, 2026.
8. From survey to exploit: systematic security evaluation of open-source space software. Bitetto, Mirko, 2025.
7. Realistic adversarial attacks against traffic sign recognition systems using stickers and graffiti. Bruzzese, Luigi, 2025.
6. Novel attack strategies targeting PROFIbus and PROFIsafe exploiting error management vulnerabilities. Alfonsi, Alessio, 2024.
5. Sandboxing in space: a feasibility study on application sandboxing for small satellites. Marra, Gabriele, 2024.
4. Linux hurt itself in its confusion! Exploiting Out-of-Memory Killer for Confusion Attacks via heuristic manipulation. Bossi, Lorenzo, 2024.
3. Exploring gradient-based evasion techniques against automotive intrusion detection systems. Cerracchio, Paolo, 2024.
2. CANPass: an extensible framework for bypassing CAN peripherals on unmodified microcontrollers. De Faveri Tron, Alvise, 2021.
1. Anomaly detection system for automotive CAN using LSTM autoencoders. Nova Valcarcel, Daniel Humberto, 2019.

Advisor of over 30 Master Theses. Co-advisor of over 20 Master Theses.
Advisor of ∼10 Master Theses for the Cyber Risk Strategy and Governance Master.

Teaching Activities

Course Instructor — Ph.D.

- Advanced Research Topics In Cyber Security 2024–2025
  Politecnico & Bocconi. ∼10 students. 30 hrs/year.

Course Instructor — Masters

- Human and Physical Aspects of Security — 2024 — Present
  Politecnico & Bocconi, CSE & Cyber Risk Strategy and Governance. ∼60 students. 50 hrs/year.
- Emerging Topics in Cybersecurity — 2024 — Present
  Bocconi, Cyber Risk Strategy and Governance Master. ∼30 students. 16 hrs/year.
- Social Engineering — 2022–2024
  Bocconi, Cyber Risk Strategy and Governance Master. ∼30 students. 16 hrs/year.

Teaching Assistant — Masters & Bachelors

- Cybersecurity Technologies, Procedures and Policies — 2023 — Present
  Politecnico & Bocconi. ∼30 students. 17 hrs/year.
- Informatica B (Fundamentals of Computer Science) — 2023–2024
  Politecnico di Milano, Ingegneria Energetica e Matematica. ∼250 students. 25 hrs/year.
- Computer Security — 2019–2023
  Politecnico di Milano, CSE. ∼200 students. 10 hrs/year.

Non-University Courses

- Cybersecurity — CEFRIEL & Politecnico di Milano, "Security Specialist" specialization degree — 2022 — Present
  ∼25 students. 110 hrs/year.
- Introduction to Industrial Cybersecurity — MADE Competence Center — 2023 — Present
  Various single-day instances (2–8 hrs).
- Cybersecurity — POLI.DESIGN, Fundamentals of the air transport system — 2019–2022
  ∼20 students. ∼10 hrs/year.

Recorded Courses

- Introduction to Industrial Cybersecurity (2024). MADE Competence Center. 4 hours of lectures.

Other Activities

- Seminars on social engineering for WindTre — 2025–2026
- Interviews on social engineering for RAI — 2024–2026
- Consulting work for RAI Radiotelevisione Italiana — 2024